

Sums

Table MOD 1

i) Single Channel Queue

No. of Cus	Inter Arrival time	Arrival Time	Service Time	Time of Service Begin	Waiting time	Time of Service End	TCS	Time Cus spends in System	ID
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$$WT = \text{Time of service Begin} - AT \geq 0$$

$$TCS = \text{Time of service End} - AT = ST + WT$$

$$\text{IDLE Time} = AT - TSE > 0$$

Formula

$$\text{Avg WT} = \frac{\text{Total WT}}{\text{no. of Cus}}, \text{ Prob of waiting} = \frac{\text{no. of WT}}{\text{no. of Cus}}, \text{ Prob of IT} = \frac{\text{Idle Time}}{TSE}$$

$$\text{Avg ST} = \frac{\text{Total ST}}{\text{no. of Cus}}, \text{ Avg AT} = \frac{\text{Total AT}}{\text{no. of Cus}}, \text{ Avg WT for those who wait} = \frac{\text{Total WT}}{\text{no. of Cus who wait}}$$

$$\text{Avg TCS} = \frac{\text{Total TCS}}{\text{no. of Cus}}, \text{ Server Utilization} = \frac{\text{Total ST}}{TSE}$$

2) Able Baker

No. of Cus	Inter AT	Arrival Time	Able			Baker			Caller Delay	Time in System
			Time SB	ST	TSE	TSB	ST	TSE		

Able - first choice if taken Baker then Baker

If both taken calc Caller delay = $\min(TSE_A, TSE_B) - AT$ wait for whichever ends first

$$\text{Time in System} = TSE + \text{Caller Delay} + ST + \text{Caller Delay}$$

$$\text{Avg wait time} = \frac{\text{Total CD}}{\text{no. of Cus}}$$

$$\text{Avg IT}_A = \frac{\text{Total IT}_A}{T \text{ Simulation Time}}, \text{ Avg IT}_B = \frac{\text{Total IT}_B}{\text{Total Sim time}}$$

$$\text{Avg ST}_A = \frac{\text{Total ST}_A}{\text{no. of Cus}}, \text{ Avg ST}_B = \frac{\text{Total ST}_B}{\text{no. of Cus}}$$

3) News Dealer

Day	Type of news day	Demand	Rev from Sales	Lost Profit	Salvage	Daily Profit
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Demand $\rightarrow$ RD + look at Type of news day

$$LP = (\text{Demand} - \text{No. of Paper}) \times (SP - CP)$$

$$\text{Salvage} = (\text{No. of Paper} - \text{Demand}) \times (\text{Scrap} \text{ Cost})$$

$$DP = \text{Rev} - \text{LP} - (\text{No. of Paper}) \times CP$$

$$\text{Daily Profit} = \text{Rev} - \text{Cost of NP} - \text{Lost P} + \text{Salvage}$$

$$\text{Rev} = \begin{cases} \text{No. of Paper} \times SP & \text{if Demand} \geq \text{No. of Paper} \\ \text{Demand} \times SP & \text{else} \end{cases}$$

4) Order up to level $\rightarrow M \rightarrow$ Order up to level, $N \rightarrow$ Cycle

Day	Cycle
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4) Order up to level $\rightarrow M \rightarrow$ Order up to level, $N \rightarrow$ Cycle

Day	Cycle No	Day within	Beginning Inv	Demand	Ending Inv	Shortage	Order Quantity	Lead time	Pay until
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$N \Rightarrow$ No. of cycle

$$\text{Ending Inv} = \text{Beginning Inv} - \text{Demand} > 0$$

$$\text{Shortage} = \text{Demand} - \text{Beginning Inv} > 0$$

$$\text{Order Quantity} = \frac{N}{M} \times \text{Ending Inv of the Cycle}$$

$$\text{Order Quantity} = M - \text{Ending Inv of the Cycle}$$

$$\text{Avg Ending Inv} = \frac{\text{Total Ending Inv}}{\text{No. of Days}}$$

5) Reliability Problem

No° of Runs	Bearing N Life (Hour)	First Failure	Delay (min)

First Failure = min (between Bearing 1, 2, ..., N)

① Cost of bearing = (No° of Runs $\times$ no° of Bearing (N)) $\times$ Cost of each bearing

② Cost of delay time = Total Delay $\times$ Cost of delay per min

③ Cost of downtime during repair = (No° of Run $\times$ N) $\times$ Change time $\times$ Cost of
downtime per min

④ Cost of mechanics / repair person = (No° of Run $\times$ N) $\times$ change time $\times$
Cost mech per hr

60

Total Cost = ① + ② + ③ + ④

Total Life of all bearing = Total First Failure $\times$ N

Total Cost per M bearing hours = $\frac{\text{Total Cost}}{\text{Total Life of all bearing}} \times M$

6) Single Queue Event Scheduling

Clock (t)	LQ(t)	LS(t)	Future Event List	B	MQ
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- a) System States $\Rightarrow$ LQ(t), LS(t)
- | | | |
|--|---|--|
| $\downarrow$
No. of cus
in Queue | $\downarrow$
No. of cus
in Server | For AT id LS(t) $\neq 1$, LQ(t) + 1
else LS(t) = 1
For DT id LS(t) = 1 & LQ(t) > 0, LS(t) = 0
else LQ(t) - 1
else LS(t) = 0 |
|--|---|--|

b) Entities : The Server & customers are not explicitly modeled, except in terms of state variables

c) Events : Arrival (A), Departure (D), Stopping Event (E)

d) Event Notices : (A, t), (D, t), (E, 60)

e) Activities : IAT, ST

f) Delay

$\rightarrow$ Cumulative Statistics

a) B $\Rightarrow$ total busy time

b) NQ $\Rightarrow$ Maximum Queue

c) S $\Rightarrow$ Sum of cus Response time for all customer who have departed by the current time

d) F $\Rightarrow$ Total no. of customer who did M or more min.

e) ND $\Rightarrow$ Total no. of Departure

f) Server Utilization = $\frac{B}{\text{Total time}}$

7) Dump truck

Clock	LQ(t)	L(t)	WQ(t)	w(t)	LQ	WQ	FEL	BL	BS
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a) System State $\Rightarrow LQ(t), L(t), WQ(t), w(t)$
 $\swarrow$ $\swarrow$ $\swarrow$ $\swarrow$
 no. of trucks in LQ no. of trucks being loaded no. of trucks in WQ no. of trucks being weighed

b) Event Notices $\Rightarrow (ALQ, t, DT), (EL, t, DT), (EW, t, DT)$

c) Entities $\Rightarrow$ No. of trucks

d) Lists $\Rightarrow LQ, WQ$
 $\swarrow$ $\swarrow$
 Loader Queue Weigh Queue

e) Activities $\Rightarrow LT, WT, TT$
 $\swarrow$ $\swarrow$ $\swarrow$
 Loading Time Weighing Time Total Time
 Travel Time

f) delay

$\rightarrow$ Cumulative Statistics

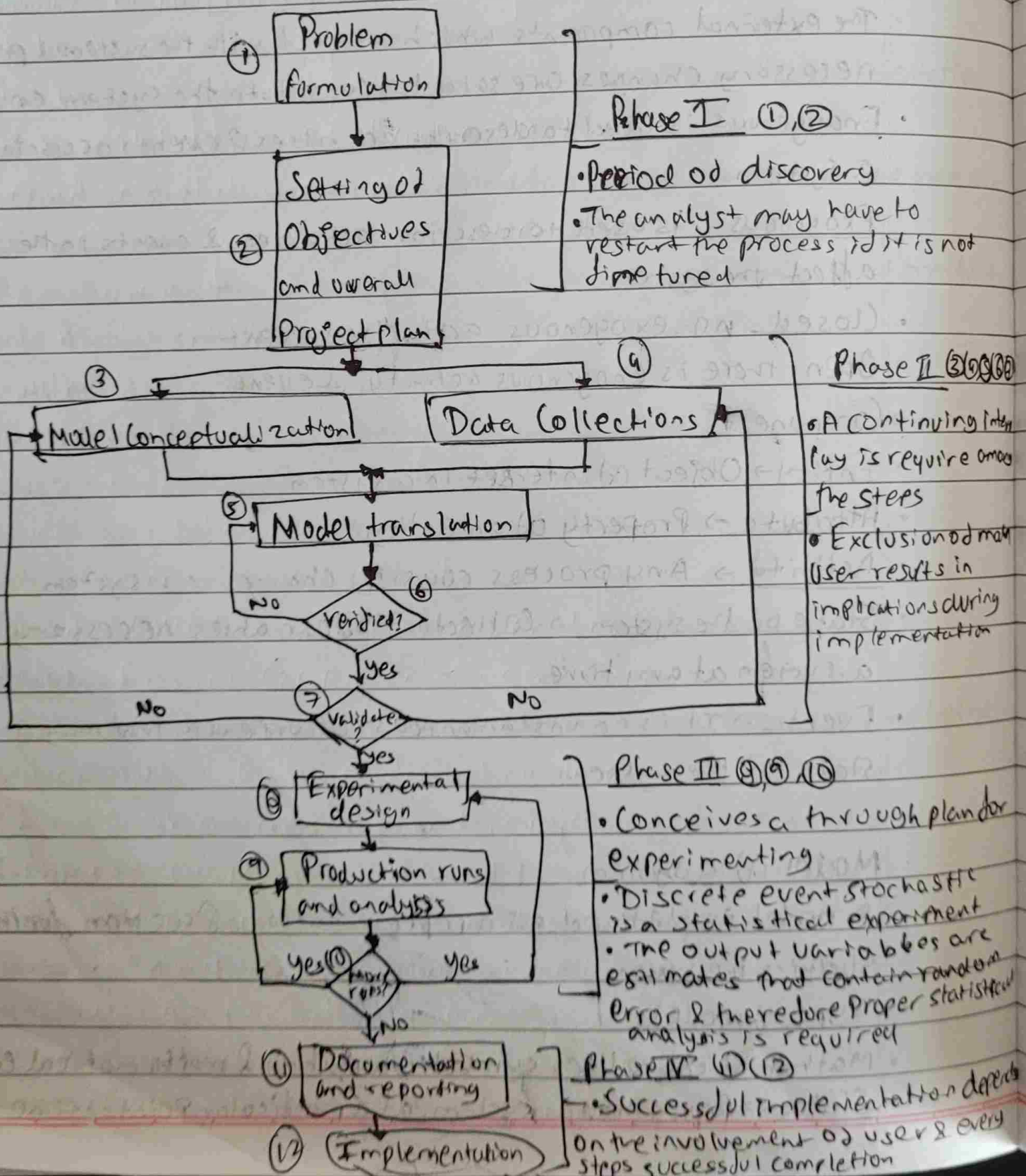
a) BL $\Rightarrow$ total busy time of both (or N) loaders

b) BS $\Rightarrow$ total busy time of scale

c) Avg Loader Utilization $= \frac{BL}{N \times t}$

d) Avg Scale Utilization $= \frac{BS}{t}$

Discrete-Event System



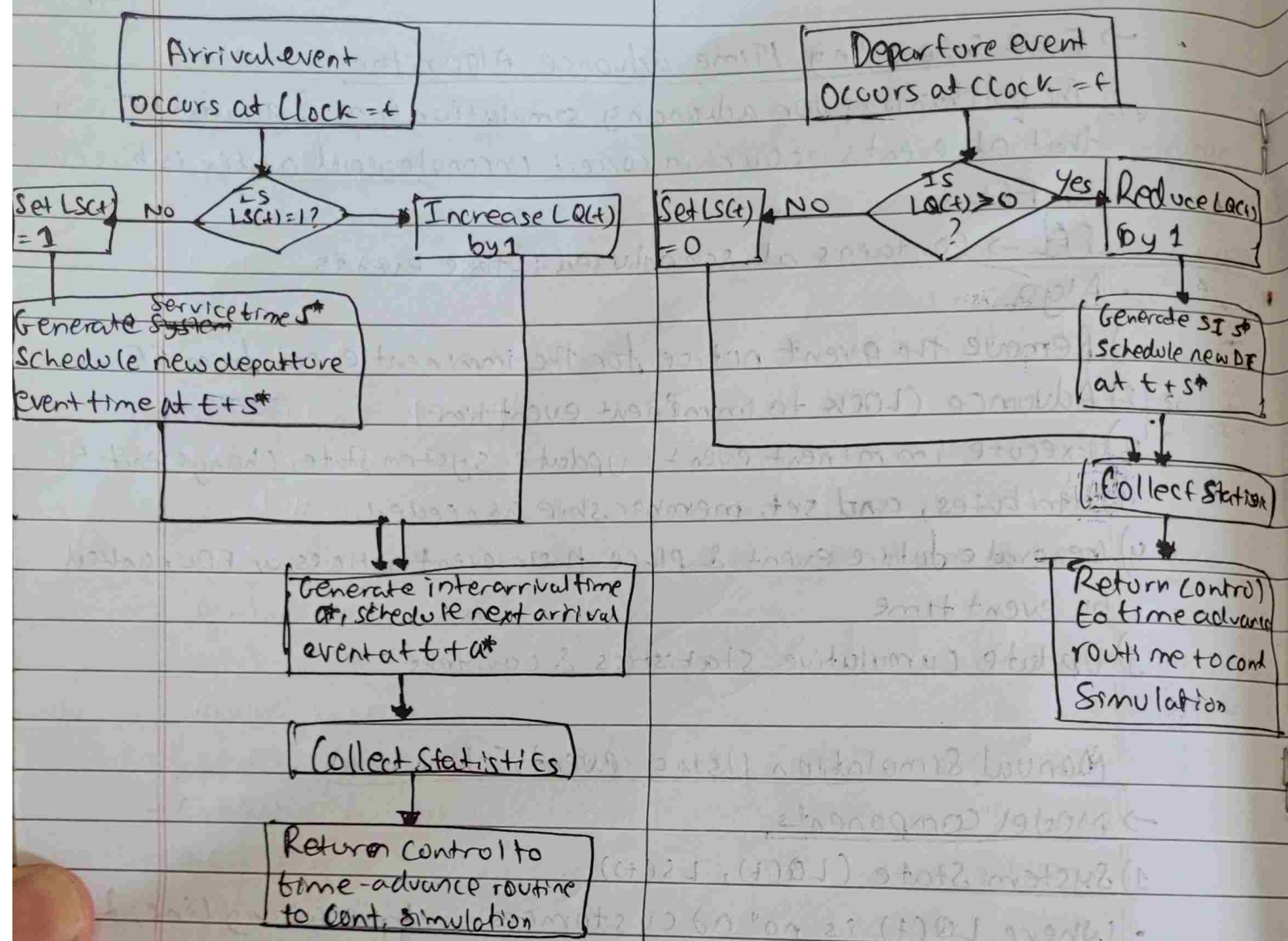
→ Event Scheduling / Time advance Algorithm

- The mechanism for advancing simulation time & guaranteeing that all events occur in correct chronological order is based on FEL

• FEL → Contains all scheduled future events.

• Algo

- 1) Remove the event notice for the imminent event from FEL
- 2) Advance CLOCK to imminent event time
- 3) Execute imminent event: update system state, change entity attributes, and set membership as needed.
- 4) Generate future event & place their event notices on FEL ranked by event time
- 5) Update cumulative statistics & counters



Mod 2-1

→ Discrete Distribution

→ Expected Value

1) Discrete $E(x) = \sum x_i P(x_i)$

2) Cont. $E(x) = \int_{-\infty}^{\infty} x f(x) dx$

→ Variance

$$V(x) = E(x^2) - [E(x)]^2$$

→ SD

$$\sigma = \sqrt{V(x)}$$

→ Coefficient of Variation

$$C.O.V = \frac{\sigma}{\mu}, \mu > 0$$

→ Covariance

$$\text{Cov}(x, y) = E(xy) - E(x)E(y) = \sigma_{xy}^2$$

→ Correlation Coeff

$$\text{Corr}(x, y) = \frac{\sigma_{xy}^2}{\sigma_x \sigma_y}$$

→ Queueing Systems → AT, ST (Distribution)

- Exponential → If ST is completely random
- Normal → fairly constant but with some random variability
- Truncated normal
- Gamma and Weibull → more general than expo.

→ Inventory & Supply chain

• Random variables

- 1) No. of units demanded per order or per time period
- 2) Time between demand
- 3) Lead time

• For lead time → Gamma

• For Demand → Poisson, negative binomial, Geometric

→ Reliability & maintainability

- Time to Failure : Exponential, Gamma, Weibull, Normal

→ Discrete

1) Bernoulli Trials & Bernoulli Distribution

- Special case of the binomial distribution

• Bernoulli Trials

$$P_j(x_j) = P(x_j) = \begin{cases} p & x_j = 1, j = 1, 2, \dots, n \\ 1-p=q & x_j = 0, j = 1, 2, \dots, n \\ 0 & \text{o.w.} \end{cases}$$

$$E(x_j) = p, V(x_j) = pq$$

• Bernoulli Distribution

$$P(x_1, x_2, \dots, x_n) = P(x_1) P(x_2) \dots P(x_n)$$

• Application

- Coin Tosses
- Pass/Fail
- Customer Behavior

2) Binomial Distribution

$$P(x) = \begin{cases} {}^n C_x p^x q^{n-x} & , x = 0, 1, \dots, n \\ 0 & , \text{o.w.} \end{cases}$$

$$E(x) = np$$

$$V(x) = npq$$

• Application

- To classify defective or non defective
- Find out demand

3) Poisson Distribution

$$P(x) = \begin{cases} \frac{e^{-\alpha} \alpha^x}{x!}, & x = 0, 1, \dots, n \\ 0, & \text{o.w} \end{cases} \quad F(x) = \sum_{i=0}^x \frac{e^{-\alpha} \alpha^i}{i!}$$

$$E(x) = V(x) = \alpha$$

→ Continuous

1) Uniform Distribution

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{o.w} \end{cases}, \quad F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x \geq b \end{cases}$$

$$P(x_1 \leq x < x_2) = \frac{x_2 - x_1}{b - a}$$

$$E(x) = \frac{a+b}{2}, \quad V(x) = \frac{(b-a)^2}{12}$$

2) Exponential Distribution

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{o.w} \end{cases}, \quad F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$

$$E(x) = \frac{1}{\lambda}, \quad V(x) = \frac{1}{\lambda^2}$$

3) Normal Distribution

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2}, \quad F(x) = \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2} \left(\frac{t-\mu}{\sigma} \right)^2} dt$$

$$= \Phi \left(\frac{x-\mu}{\sigma} \right)$$

n) Erlang Distribution

$$f(x) = \begin{cases} \frac{\beta^\theta}{\Gamma(\theta)} (\beta x)^{\theta-1} e^{-\beta x}, & x > 0, \beta = k \text{ or int} \\ 0, & \text{o.w} \end{cases}$$

$$F(x) = \begin{cases} 1 - \sum_{i=0}^{k-1} \frac{e^{-kx} (kx)^i}{i!}, & x > 0 \\ 0, & \text{o.w} \end{cases}$$

$$E(x) = 1/\theta, V(x) = 1/k\theta^2, \text{Mode} = \frac{k-1}{k\theta}$$

Mod 2.2

→ M/M/1
↑
Poisson Exponential

• Poisson Arrival (M) → Arrival $\Rightarrow \lambda$ ← average no. of arrival per time unit
Inter-Arrival $\Rightarrow$ Exponential distributed mean $= 1/\lambda$

• Exponential Service Time (M) $\Rightarrow 1/\mu$ ← service rate

Formulae

1) Utilization Factor (ρ) $= \frac{\lambda}{\mu}$

2) Average no. of Customers in the system $= L = \frac{\lambda}{\mu - \lambda}$

3) " " " " " " Queue $= L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} \Rightarrow \frac{\lambda}{\mu} L$

4) " Time " " spends in the System $= W = \frac{1}{\mu - \lambda}$

5) " " " " " " Queue $= W_q = \frac{\lambda}{\mu(\mu - \lambda)} \Rightarrow \frac{\lambda}{\mu} W$

6) Probability of having n Customers in the System $= P_n = (1 - \rho) \rho^n$

• Application

- Telecommunication
- Computer Networks

→ M/G/1

↑ ↑
Poisson General

• Formula

$$1) \rho = \frac{\lambda}{\mu}$$

$$2) L = \frac{\lambda}{\mu} + \frac{\lambda^2 \sigma^2}{2\mu(1-\rho)} = \frac{\lambda}{\mu} + L_q$$

$$3) L_q = \frac{\lambda^2 \sigma^2}{2\mu(1-\rho)}$$

$$4) W = \frac{1}{\mu} + \frac{\lambda \sigma^2}{2\mu(1-\rho)} = \frac{1}{\mu} + W_q$$

$$5) W_q = \frac{\lambda \sigma^2}{2\mu(1-\rho)}$$

• Application

- Telecommunication
- Customer service

(3.3)

(I) Inverse

Algo

- 1) Get the CDF of the desired Random Variable x
- 2) Generate $R \sim U(0,1)$ & set $F(x) = R$ on the range of x
- 3) Calculate F^{-1} by solving the equation $F(x) = R$
- 4) Generate uniform random no. $x = F^{-1}(R)$

Graphical Method

- 1) Plot the CDF
- 2) For generating a random variable x with CDF, $F(x)$
- 3) Draw a horizontal line through vertical axis at $F(x) = R$ so that it intersects the CDF
- 4) Vertical line dropped from that point.

1) Exponential

$$CDF = F(x) = 1 - e^{-\lambda x}$$

$$R = 1 - e^{-\lambda x}$$

$$e^{-\lambda x} = 1 - R$$

$$-\lambda x = \ln(1 - R)$$

$$x = \frac{-1}{\lambda} \ln(1 - R)$$

$$\underline{x = -\frac{1}{\lambda} \ln(1 - R)}$$

2) Uniform Distribution

$$F(x) = 0 \quad x < a$$

$$= \frac{x - a}{b - a} \quad a \leq x \leq b$$

$$= 1 \quad x > b$$

$$= 1 \quad x > b$$

$$\frac{x - a}{b - a} = R$$

$$R(b - a) = x - a$$

$$x = a + R(b - a)$$

3) Weibull Distribution

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-(x/\alpha)^\beta} & x \geq 0 \end{cases}$$

$$X \geq 1 - e^{-(x/\alpha)^\beta} = R$$

$$R | - R = e^{-(x/\alpha)^\beta}$$

$$-\ln(1-R) = -\left(\frac{x}{\alpha}\right)^\beta$$

$$x = -\alpha [\ln(1-R)]^{1/\beta}$$

4) Triangular

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \frac{x^2}{c} & 0 < x \leq 1 \\ \frac{1 - (c-x)^2}{c(c-1)} & 1 < x \leq c \\ 1 & x > c \end{cases}$$

$$\frac{x^2}{c} = R \Rightarrow x = \pm \sqrt{cR} \Rightarrow 0 < R \leq 1/c$$

$$\frac{1 - (c-x)^2}{c(c-1)} = R$$

$$c(c-1)(1-R) = (c-x)^2 \Rightarrow \frac{1}{c} \leq R \leq 1$$

$$c \pm \sqrt{c(c-1)(1-R)} = x$$

Empirical Cont Distribution

Algo

- 1) Sort the data point in increasing order
- 2) Make this table

Interval	Prob ($1/n$)	Cum Prob (i/n)	Slope a_i
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- 3) Draw the observed empirical CDF & follow the Graphical method
- 4) Compute $a_i = n(x_i - x_{i-1})$
- 5) Get the CDF of the desired R.V. n

$$F(x) = \begin{cases} 0 & x < x_1 \\ \frac{i-1}{n} + \frac{x - x_{i-1}}{a_i} & x_{i-1} \leq x < x_i \\ 1 & x \geq x_n \end{cases}$$

- 6) Do Inverse transform

$$\frac{i-1}{n} + \frac{x - x_{i-1}}{a_i} = R$$

$$\frac{x - x_{i-1}}{a_i} = R - \left(\frac{i-1}{n}\right)$$

$$x = x_{i-1} + a_i \left(R - \left(\frac{i-1}{n}\right) \right)$$

Discrete

Uniform

$$P(x) = 1/k, \quad x = 1, 2, \dots, k$$

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/k & 1 \leq x < 2 \\ \frac{i-1}{k} & i-1 \leq x < i \\ 1 & x \geq k \end{cases}$$

$$r_i = F(x_i) \Rightarrow r_i = \frac{i}{k} \quad \therefore \frac{i-1}{k} < R \leq \frac{i}{k}$$

$$(i-1)/k < R \leq i/k \quad \therefore Rk \leq i < Rk+1$$

$$X \sim [R, K]$$

$$1^R(n-1)$$

Geometric

$$F(x) = 1 - (1-p)^{n+1}$$

$$\therefore 1 - (1-p)^n < R < 1 - (1-p)^{n+1}$$

$$(1-p)^{n+1} \leq \frac{1-R}{p} < (1-p)^n$$

$$(n+1) \ln(1-p) \leq \ln(1-R) \leq n \ln(1-p)$$

$$n+1 \leq$$

$$\frac{\ln(1-R) - 1}{\ln(1-p)} \leq n < \frac{\ln(1-R)}{\ln(1-p)}$$

$$\frac{\ln(1-R) - 1}{\ln(1-p)}$$

$$\therefore X = \left\lceil \frac{\ln(1-R) - 1}{\ln(1-p)} \right\rceil \Rightarrow \left\lceil \frac{\ln(1-R) - 1}{\ln(1-p)} \right\rceil$$

II) Convolution Method

Erlang Distribution

We know, $X = \sum_{i=1}^K X_i$ & Exponential

$$X_i = -\frac{1}{\lambda} \ln(1-R_i) = -\frac{1}{\lambda} \ln(R_i)$$

$$X = \sum_{i=1}^K -\frac{1}{\lambda} \ln(R_i)$$

$$\text{We know } \frac{1}{\lambda} = \frac{1}{K\theta}$$

$$\therefore X = -\frac{1}{K\theta} \sum_{i=1}^K \ln(R_i) = -\frac{1}{K\theta} \ln \left[\prod_{i=1}^K R_i \right]$$

IV) Acceptance Rejection Technique

Algo

- 1) Generate Random no' R
- 2a) if $R \geq 1/4$ accept $x = R$ goto step 3
- 2b) if $R < 1/4$ reject R , return to step 1
- 3) if another uniform random variate on $[1/4, 1]$ is needed repeat ~~the~~ step 1, if not, stop

Formula

~~LCG~~ LCG

$$LCG \Rightarrow X_{i+1} = (aX_i + c) \bmod m \quad R_i = \frac{X_i}{m}$$

$\downarrow$
 $c = 0 \text{ mcm}$

$$P = m = 2^b \quad \text{if } c \neq 0 \text{ \& } a = 1 + 4k \text{ \& } \gcd(c, m) = 1$$

$$P = \frac{m}{4} = 2^{b-2} \quad \text{if } c = 0 \text{ \& } x_0 \text{ is odd \& } a = 3 + 8k \text{ or } 5 + 8k$$

$$P = m - 1 \quad \text{if } c = 0 \text{ \& } m \text{ is prime \& } a^{m-1} \text{ divisible by } m \text{ \& } k = m - 1$$

CLCG

$$X_i = \left(\sum_{j=1}^k (-1)^{j+1} X_{i,j} \right) \bmod m_i - 1 \quad R_i = \frac{X_i}{m_i} \quad X_i > 0$$

$$\frac{m_i - 1}{m_i} \quad X_i \neq 0$$

$$P = \frac{(m_1 - 1) \cdots (m_k - 1)}{2^{k-1}}$$

Two LCG

$$X_i = (X_{1,i} - X_{2,i}) \bmod (m_i - 1)$$

Tests

i) Frequency $\Rightarrow H_0: R_i \sim U[0,1]$
 $H_1: R_i \sim U[0,1]$

ii) KS Test

$$D^+ = \max \left\{ \frac{i}{N} - R_i \right\}, D^- = \max \left\{ R_i - \frac{i-1}{N} \right\} \quad N \leq 50$$

$$\max(D^+, D^-) = D$$

$\therefore D > D_\alpha$ H_0 is rejected

ii) Chi Square test $N \geq 30$

$$\chi_o^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad E_i = \frac{N}{n}$$

id $\chi_o^2 > \chi_{\alpha, N-1}^2 \therefore H_0$ is Rejected

2) Auto correlation Test $\Rightarrow H_0 = R_i \overset{N}{\text{Independent}}$
 $H_1 = R_i \nmid \text{Independent}$

$$\hat{\rho}_{im} = \frac{1}{M+1} \left[\sum_{k=0}^M R_{i+km} R_{i+(k+1)m} \right] - 0.25 \quad Z_0 = \frac{\hat{\rho}_{im}}{\hat{\sigma}_{\rho_{im}}}$$

$$\hat{\sigma}_{\rho_{im}} = \frac{\sqrt{13M+7}}{12(M+1)}$$

$$i + (M+1)m \leq N$$

$$-Z_{\alpha/2} \leq Z_0 \leq Z_{\alpha/2}$$

Runs UP & RUNS Down

$$M_a = \frac{2N-1}{3}, \quad \sigma_a^2 = \frac{16N-29}{90}$$

$$Z_0 = \frac{0 - M_a}{\sigma_a}$$

Where $a = \text{Total RUNS}$

$$\begin{array}{cccc} 0.12 & 0.22 & 0.1 & 0.5 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ + & + & - & + \\ \text{(1)} & \text{(2)} & & \end{array}$$

$$++ = ++ =$$

Runs Above & below means (0.495 or 49.5)

$$M_b = \frac{2n_1 n_2}{N} + \frac{1}{2}, \quad \sigma_b^2 = \frac{2n_1 n_2 (2n_1 n_2 - N)}{N^2 (N-1)}, \quad Z_0 = \frac{b - M_b}{\sigma_b}$$

Acceptance Rejection on Poisson

- 1) Set $n \geq 0$, $p = 1$
- 2) Generate Random no. R_{n+1} & replace $p = p \cdot R_{n+1}$
- 3) if $p \leq e^{-n}$ then accept $N = n$
ow. reject current n & increase it by 1
& return to step 2

Solve

MODY

Formulas

→ Grouped Data

$$\bar{X} = \frac{\sum f_i x_i}{n}$$

$$S^2 = \frac{\sum f_i x_i^2 - n(\bar{x})^2}{n-1}$$

→ UnGrouped Data

$$\bar{X} = \frac{\sum x_i}{n}$$

$$S^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$$

MLE

$$\text{Poisson} \Rightarrow \hat{\lambda} = \bar{X}$$

$$\text{Expo} \Rightarrow \hat{\lambda} = 1/\bar{x}$$

$$\text{Normal} \Rightarrow \hat{\mu} = \bar{x}, \hat{\sigma}^2 = s^2$$

Goodness of fit

CHI square $\rightarrow n \geq 50$ no. of obs no. of parameters

$$\chi_o^2 = \sum_{E_i} (O_i - E_i)^2, \text{df} = K - J - 1$$

KS Test

$$D = \max |F(x) - S_n(x)|$$

Multivariate

~~Cov(X_1, X_2)~~

$$\text{Cov}(X_1, X_2) = \frac{1}{n-1} \sum_{j=1}^n (X_{1j} - \bar{X}_1)(X_{2j} - \bar{X}_2)$$

$$\text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sigma_1 \sigma_2}$$

Time Series

$$\rho_h = \text{corr}(X_t, X_{t+h}) = \phi_h = \frac{\text{Cov}(X_t, X_{t+h})}{\hat{\sigma}_2}$$

AR(1)

$$X_t = \mu + \phi(X_{t-1} - \mu) + \epsilon_t, \quad t=2, 3, \dots \quad |\phi| < 1$$

$$\mu = \bar{X}, \quad \hat{\sigma}^2(1 - \phi^2) = \hat{\sigma}_\epsilon^2$$

EAR(1)

$$X_t = \begin{cases} \phi X_{t-1} & \text{with } \phi \\ \phi X_{t-1} + \epsilon_t & \text{with } 1 - \phi \end{cases}$$

$$\hat{\lambda} = \frac{1}{\bar{X}}$$